Neap

VCE Specialist Mathematics Units 3&4

Suggested Solutions

2025 Trial Examination 2

Section A – Multiple-choice questions

1	A	**B**	C	D
2	A	B	**C**	D
3	A	B	C	**D**
4	**A**	B	C	D
5	A	**B**	C	D
6	A	B	C	**D**
7	A	B	**C**	D
8	A	B	**C**	D
9	A	**B**	C	D
10	A	B	**C**	D
11	A	**B**	C	D
12	**A**	B	C	D
13	A	B	C	**D**
14	A	B	**C**	D
15	A	**B**	C	D

16	A	B	C	**D**
17	**A**	B	C	D
18	A	B	C	**D**
19	A	**B**	C	D
20	**A**	B	C	D

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Question 1 B

The contrapositive of a statement is negated and reversed. Therefore, the contrapositive of the statement, 'if m is odd, then $m^2 + 3m - 4$ is even', is 'if $m^2 + 3m - 4$ is odd, then m is even'.

Question 2 C

$$f'(x) = \frac{x^2 + 2x + p - 1}{(x+1)^2}$$

The equation $f'(x) = 0$ will have no solutions when the discriminant of $x^2 + 2x + p - 1 = 0$ is less than zero.

Therefore:

$$2^2 - 4 \times 1 \times (p-1) < 0$$
$$p > 2$$

The function will have no turning points when $p > 2$.

Question 3 D

$$z = 100\sqrt{2}\,\text{cis}\left(\frac{\pi}{4}\right)$$

$$z^{n+1} = \left(100\sqrt{2}\right)^{n+1}\text{cis}\left((n+1)\frac{\pi}{4}\right)$$

For z^{n+1} to be a real number, its argument must be an integer multiple of π.

Therefore:

$$(n+1)\frac{\pi}{4} = \pi k$$
$$(n+1)\pi = 4\pi k$$
$$n + 1 = 4k$$
$$n = 4k - 1,\ k \in Z$$

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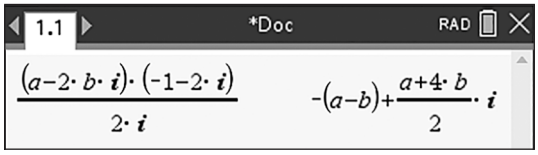
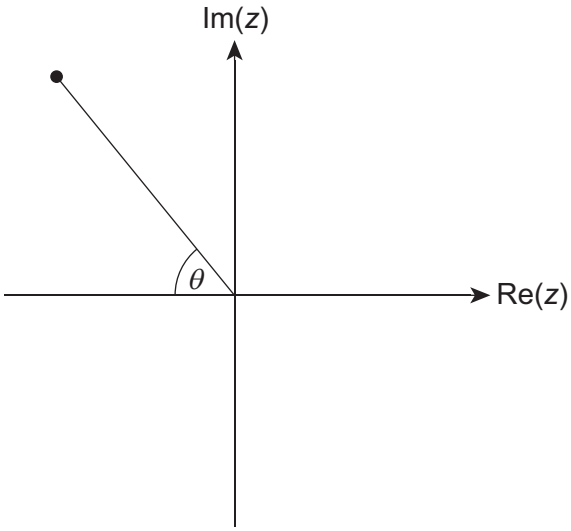
Question 4 A

Using a CAS calculator gives:



$$\frac{\bar{z}w}{2i} = -(a-b) + \frac{a+4b}{2}i$$

As $a > b > 0$, $\frac{\bar{z}w}{2i}$ is in the second quadrant.



$$\theta = \tan^{-1}\left(\frac{\frac{a+4b}{2}}{a-b}\right)$$

$$= \tan^{-1}\left(\frac{a+4b}{2(a-b)}\right)$$

$$\therefore \text{Arg}\left(\frac{\bar{z}w}{2i}\right) = \pi - \tan^{-1}\left(\frac{a+4b}{2(a-b)}\right)$$

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Question 5 B

Using a CAS calculator to find the approximate value of y_5 gives:

$$\text{euler}\left(e^{x-y}\cdot\sin(x),x,y,\left\{\frac{\pi}{2},\frac{\pi}{2}+0.5\right\},\frac{\pi}{2},0.1\right)$$

$$\begin{bmatrix} 33 & 1.87079633 & 1.97079633 & 2.07079633 \\ 74 & 1.86835238 & 1.96411979 & 2.0568429 \end{bmatrix}$$

$\therefore\ y_5 = 2.056\ \ldots$

Using a CAS calculator to find the exact solution to the differential equation gives:

$$\text{deSolve}\left(y'=e^{x-y}\cdot\sin(x)\text{ and }y\left(\frac{\pi}{2}\right)=\frac{\pi}{2},x,y\right)$$

$$e^{y}-e^{\frac{\pi}{2}}=\frac{-\sqrt{2}\cdot e^{x}\cdot\cos\left(x+\frac{\pi}{4}\right)}{2}-\frac{e^{\frac{\pi}{2}}}{2}$$

$$e^{y}-e^{\frac{\pi}{2}}=-\frac{\sqrt{2}}{2}\cos\left(x+\frac{\pi}{4}\right)-\frac{1}{2}e^{\frac{\pi}{2}}$$

When $x=\frac{\pi}{2}+0.5,\ y=2.052\ \ldots.$

Therefore, the difference is closest to 2.056 … – 2.052 … = 0.004.

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Question 6 D

Using similar triangles gives:

$$\frac{r}{h} = \frac{30}{50}$$

$$h = \frac{5r}{3}$$

Using the chain rule gives:

$$\frac{dr}{dt} = \frac{dr}{dV}\frac{dV}{dt}$$

Finding the initial volume of water gives:

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi r^2\left(\frac{5r}{3}\right) \\ &= \frac{5\pi r^3}{9} \end{aligned}$$

Therefore, $\frac{dV}{dr} = \frac{5\pi r^2}{3}$.

Given that $\frac{dV}{dt} = -\frac{k\sqrt{h}}{2}$:

$$\frac{dr}{dt} = \frac{3}{5\pi r^2} \times -\frac{k\sqrt{h}}{2}$$

When $h = 20$, $r = 12$.

$$\begin{aligned} \frac{dr}{dt} &= \frac{3}{5\pi \times 12^2} \times -\frac{k\sqrt{20}}{2} \\ &= -\frac{k\sqrt{5}}{240\pi} \end{aligned}$$

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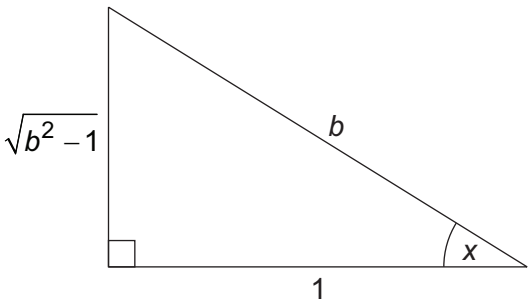
Question 7 C

If $x \in [-\pi, 0]$ and $b \in R^+$, $\cos(x)$ is negative; therefore, x is in the third quadrant.

Using a diagram to find $\sin(x)$ gives:



As x is in the third quadrant, $\sin(x)$ must be negative.

$$\sin(x) = -\frac{\sqrt{b^2-1}}{b}$$

$$\begin{aligned}
\operatorname{cosec}(2x) &= \frac{1}{\sin(2x)} \\
&= \frac{1}{2\sin(x)\cos(x)} \\
&= \frac{1}{2 \times -\frac{\sqrt{b^2-1}}{b} \times -\frac{1}{b}} \\
&= \frac{1}{\frac{2\sqrt{b^2-1}}{b^2}} \\
&= \frac{b^2}{2\sqrt{b^2-1}}
\end{aligned}$$

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Question 8 C

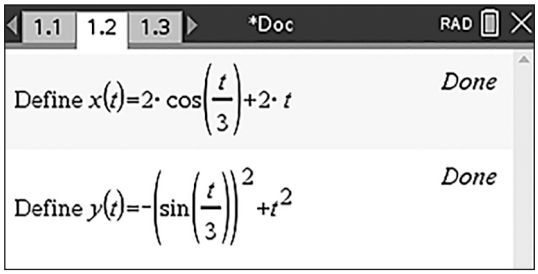
Letting $u = \log_e(nx)$ and $\frac{dv}{dx} = x^{n^2}$ gives:

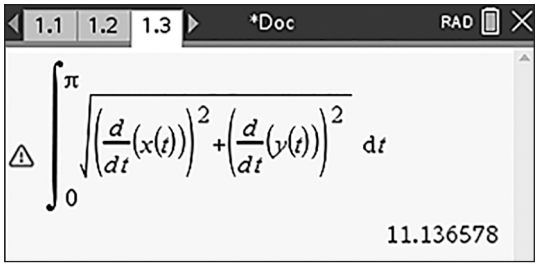
$\frac{du}{dx} = \frac{1}{x}$ and $v = \frac{x^{n^2+1}}{n^2+1}$

$$\begin{aligned}\int x^{n^2}\log_e(nx)\,dx &= \frac{x^{n^2+1}}{n^2+1}\log_e(nx) - \int \frac{1}{x}\cdot\frac{x^{n^2+1}}{n^2+1}\,dx \\ &= \frac{x^{n^2+1}}{n^2+1}\log_e(nx) - \int \frac{x^{n^2}}{n^2+1}\,dx \\ &= \frac{x^{n^2+1}}{n^2+1}\log_e(nx) - \frac{x^{n^2+1}}{(n^2+1)^2} + c \\ &= \frac{x^{n^2+1}}{n^2+1}\left(\log_e(nx) - \frac{1}{n^2+1}\right) + c\end{aligned}$$

Question 9 B

B is correct. Using a CAS calculator gives:





$\therefore$ length $= 11.14$

A is incorrect. This option does not square the derivatives under the square root.

C is incorrect. This option uses 2π, instead of π, as the upper limit of integration.

D is incorrect. This option does not include the square root in the calculation.

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Question 10 C

Finding an expression for the projectile's displacement from sea level gives:

$$\frac{dv}{dt} = -\frac{v}{20}$$

$$\frac{dt}{dv} = -\frac{20}{v}$$

$$t = \int -\frac{20}{v}\,dv$$

$$= -20\log_e|v| + c$$

Let downwards be positive.

At $t = 0$, $v = 30$.

$$0 = -20\log_e|30| + c$$

$$c = 20\log_e(30)$$

$$t = -20\log_e|v| + 20\log_e(30)$$

$$t = 20\log_e\left|\frac{30}{v}\right|$$

$$\left|\frac{30}{v}\right| = e^{\frac{t}{20}}$$

$$v = 30e^{-\frac{t}{20}}$$

This satisfies the boundary condition.

$$\frac{dx}{dt} = 30e^{-\frac{t}{20}}$$

$$x = -600e^{-\frac{t}{20}} + c_2$$

At $t = 0$, $x = 0$.

$$0 = -600e^0 + c_2$$

$$c_2 = 600$$

$$x = 600 - 600e^{-\frac{t}{20}}$$

Finding the ball's displacement from sea level gives:

initial velocity $= 0$, $a = 3$

$$x = ut + \frac{1}{2}at^2$$

$$= \frac{3}{2}t^2$$

Solving for t when the displacement of the projectile is five times that of the ball gives:

$$600 - 600e^{-\frac{t}{20}} = 5 \times \frac{3}{2}t^2$$

$$t = 3.65\ldots$$

At $t = 3.65\ldots$, the projectile is 100.23 m deep, and the ball is 20.05 m deep. Therefore, the distance between them is closest to 80 m.

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Question 11 B

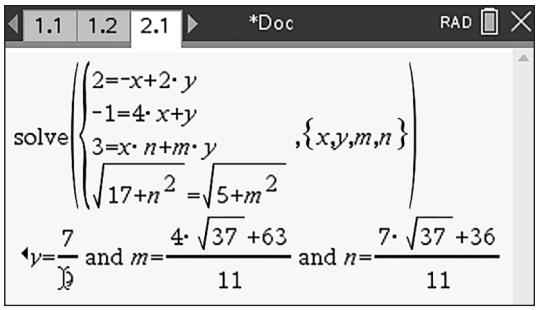
Using a definite integral gives:

$$\frac{dx}{dt} = 2\cos\left(t^2\right)\sin\left(t^2\right)$$

$$\begin{aligned} x &= \int_0^2 2\cos\left(t^2\right)\sin\left(t^2\right)dt \\ &= 0.4537\ldots \\ &= 0.45\text{ m} \end{aligned}$$

Question 12 A

Using a CAS calculator gives:



$$\therefore m = \frac{4\sqrt{37}+63}{11},\ n = \frac{7\sqrt{37}+36}{11}$$

Question 13 D

The coordinates of point X are (3, 0, 0); therefore, a vector parallel to the line through P and X is $\underset{\sim}{d} = -4\underset{\sim}{j} - \underset{\sim}{k}$.

The vector equation of the line can be formed using $\underset{\sim}{r}(t) = \underset{\sim}{a} + \underset{\sim}{d}t$.

Therefore, the vector equation could be:

$$\begin{aligned} \underset{\sim}{r}(t) &= 3\underset{\sim}{i} - 4\underset{\sim}{j} - \underset{\sim}{k} + t\left(-4\underset{\sim}{j} - \underset{\sim}{k}\right) \\ &= 3\underset{\sim}{i} - 4\underset{\sim}{j} - \underset{\sim}{k} + t\left(4\underset{\sim}{j} + \underset{\sim}{k}\right) \end{aligned}$$

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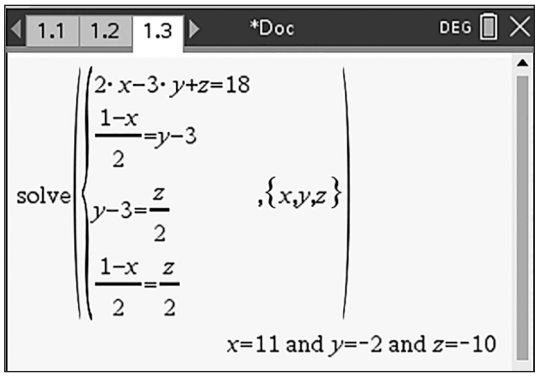
Question 14 C

Using a CAS calculator to find point *X* gives:

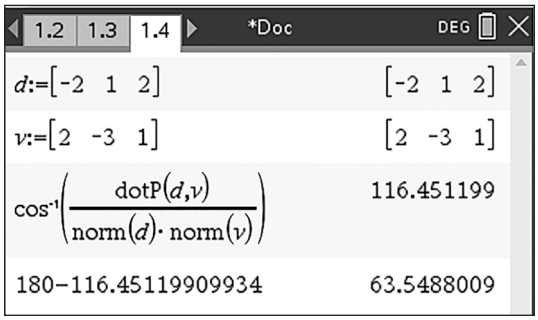


Method 1:

A vector parallel to the line is given by $\underset{\sim}{d} = -2\underset{\sim}{i} + \underset{\sim}{j} + 2\underset{\sim}{k}$.

The vector normal to the plane is given by $\underset{\sim}{v} = 2\underset{\sim}{i} - 3\underset{\sim}{j} + \underset{\sim}{k}$.

Using a CAS calculator to find the angle between the vector normal to the plane and line *L* gives:



Finding the length of *AX* using points *A* and *X* gives:

$\sqrt{(11-1)^2 + (-2-3)^2 + (-10-0)^2} = 15$

Finding the other two sides of the triangle gives:

$15\cos(63.54\ \ldots) = 6.68\ \ldots$

$15\sin(63.54\ \ldots) = 13.42\ \ldots$

Therefore, the area of the triangle is $\frac{1}{2} \times 6.68\ \ldots \times 13.42\ \ldots = 44.86\ \ldots = 45$.

(continues on next page)

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(continued)

Method 2:

Given that the triangle is a right-angled triangle, the side of the triangle that is not contained within Π represents the shortest distance between point A and the plane.

A vector normal to the plane is $\hat{\underset{\sim}{n}} = 2\underset{\sim}{i} - 3\underset{\sim}{j} + \underset{\sim}{k}$; therefore, a unit vector normal to the plane is $\hat{\underset{\sim}{n}} = \frac{1}{\sqrt{14}}\left(2\underset{\sim}{i} - 3\underset{\sim}{j} + \underset{\sim}{k}\right)$.

A vector joining another point on the plane, Q, to point A is $\overrightarrow{AQ} = (x-1)\underset{\sim}{i} + (y-3)\underset{\sim}{j} + z\underset{\sim}{k}$.

Finding the shortest distance between point A and Π gives:

$$\begin{aligned}\left|\overrightarrow{AQ}\cdot\hat{\underset{\sim}{n}}\right| &= \frac{1}{\sqrt{14}}\left(2(x-1) - 3(y-3) + z\right)\\ &= \frac{1}{\sqrt{14}}(2x - 3y + z - 2 + 9)\\ &= \frac{1}{\sqrt{14}}(18 + 7)\\ &= \frac{25}{\sqrt{14}}\\ &= 6.68\ ...\end{aligned}$$

Finding the hypotenuse using points A and X gives:

$$\sqrt{(11-1)^2 + (-2-3)^2 + (-10-0)^2} = 15$$

Finding the side of the triangle contained within Π gives:

$$\sqrt{15^2 - 6.68\ldots^2} = 13.42\ldots$$

Therefore, the area of the triangle is $\frac{1}{2} \times 6.68\ldots \times 13.42\ldots = 44.86\ldots = 45$.

Question 15 B

$$\begin{aligned}\underset{\sim}{v} \times \underset{\sim}{w} &= \left(m\underset{\sim}{a} + n\underset{\sim}{b}\right) \times \left(-2\underset{\sim}{a} + 3\underset{\sim}{b}\right)\\ &= m\underset{\sim}{a} \times -2\underset{\sim}{a} + m\underset{\sim}{a} \times 3\underset{\sim}{b} + n\underset{\sim}{b} \times -2\underset{\sim}{a} + n\underset{\sim}{b} \times 3\underset{\sim}{b}\\ &= \underset{\sim}{0} + 3m\left(\underset{\sim}{a} \times \underset{\sim}{b}\right) + -2n\left(\underset{\sim}{b} \times \underset{\sim}{a}\right) + \underset{\sim}{0}\\ &= (3m + 2n)\left(\underset{\sim}{a} \times \underset{\sim}{b}\right)\end{aligned}$$

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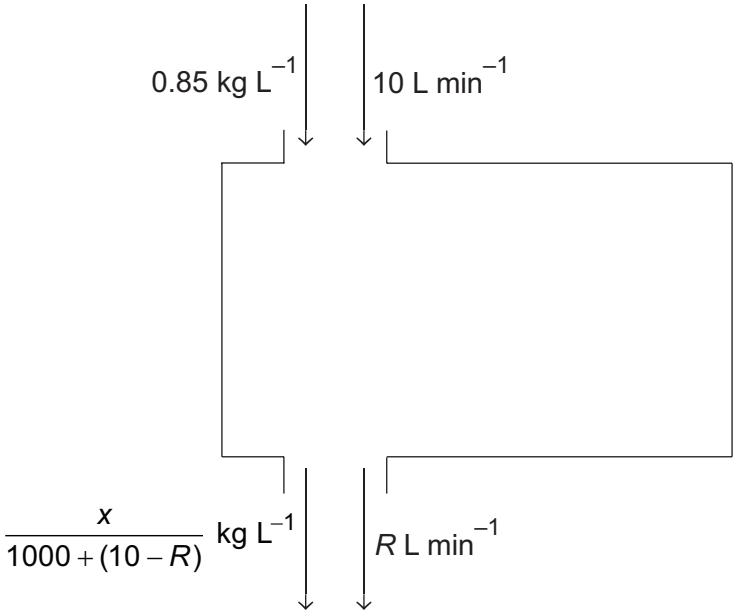
Question 16 D

The diagram below shows the rates of inflow and outflow.



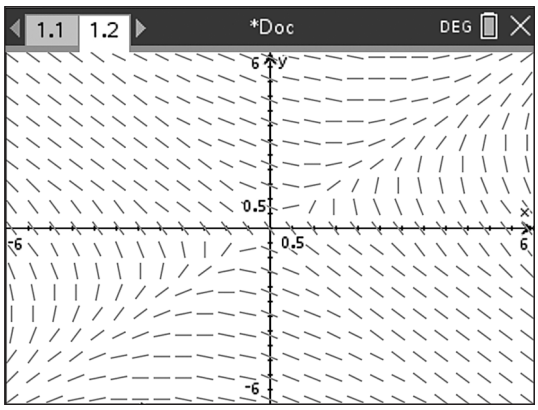
$$\frac{dx}{dt} = \text{rate of inflow} - \text{rate of outflow}$$

$$= 0.85 \times 10 - \frac{x}{1000+(10-R)} \times R$$

$$= 8.5 - \frac{Rx}{1010-R}$$

Question 17 A

A is correct. Using a CAS calculator to verify the slope field of $\frac{dy}{dx} = \frac{2x-y}{2y-x}$ gives:



B is incorrect. The gradient of this equation would be undefined when $x = \frac{1}{2}$ and $y = 1$.

C is incorrect. The gradient of this equation would be zero when $x = 1$ and $y = \frac{1}{2}$.

D is incorrect. This equation can be verified as incorrect using a CAS calculator.

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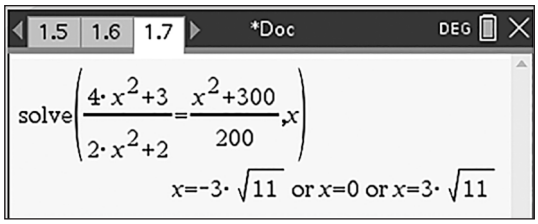
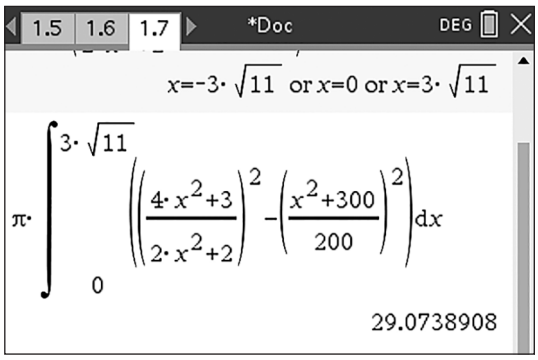
Question 18 D

Using a CAS calculator to find the x-values of the points of intersection of T and B gives:



As $x \geq 0$, $x = 0, 3\sqrt{11}$.

Using a CAS calculator to find the volume of the region bounded by T and B gives:



$\therefore\ V = 29.07$

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Question 19 B

Method 1:

Calculating $\Pr\left(\bar{X} < 97.9 \mid \mu = 99\right)$ for each option gives:

A: $n = 67$ gives 0.0508.

B: $n = 68$ gives 0.0495.

C: $n = 69$ gives 0.0483.

D: $n = 70$ gives 0.0471.

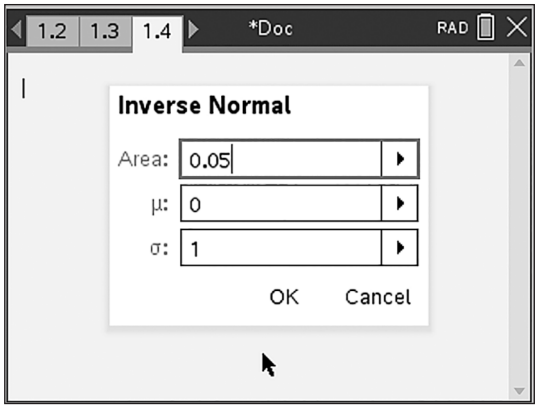
Therefore, $n = 68$ is the minimum sample size that yields a p value less than 0.05.

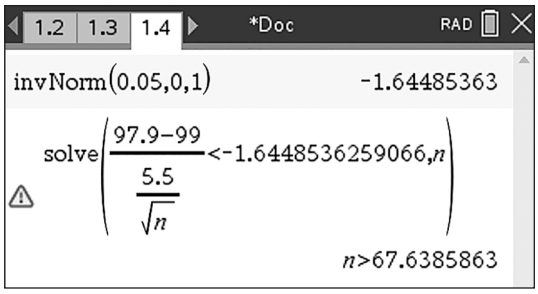
Method 2:

$$\Pr\left(\bar{X} < 97.9 \mid \mu = 99\right) < 0.05$$

$$\Pr\left(Z < \frac{97.9 - 99}{\frac{5.5}{\sqrt{n}}}\right) < 0.05$$

Using a CAS calculator to find the inverse normal distribution gives:





$$\frac{97.9 - 99}{\frac{5.5}{\sqrt{n}}} < -1.64\ldots$$

$$n > 67.63\ldots$$

Therefore, $n = 68$ is the minimum sample size.

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Question 20 A

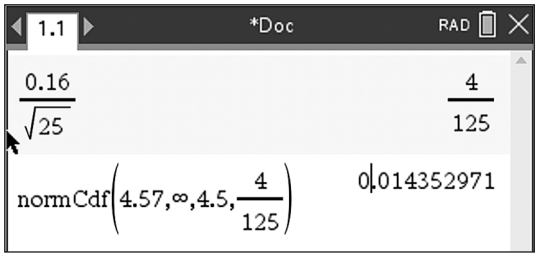
Performing a one-sided statistical test gives:

H_0: $\mu = 4.5$kg

H_1: $\mu > 4.5$ kg

p value $= \Pr\left(\bar{X} > 4.57 \mid \mu = 4.5\right)$, where $sd\left(\bar{X}\right) = \dfrac{0.16}{\sqrt{25}} = \dfrac{4}{125}$

Using a CAS calculator gives:



Therefore, the p value is 0.014 ….

As the p value is greater than the level of statistical significance (0.01), the null hypothesis should not be rejected. This means that the food does not affect wool mass.

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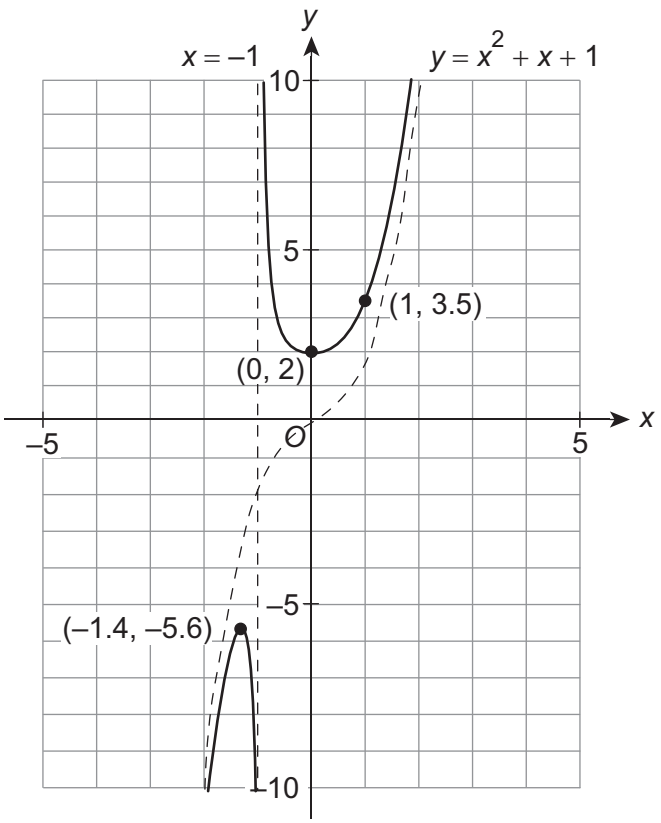
Section B

Question 1 (10 marks)

a.



correct shape A1
correct asymptotes A1
correct stationary points and point of discontinuity A1

b. **i.** $f'(x) = \dfrac{x\left(3x^3 + 6x^2 + 4x + 2\right)}{(x+1)^2}$

Finding the definite integral gives:

$$2\pi \int_{\frac{3}{2}}^{2} \left(\frac{x^5 + x^2 - 2}{x^2 - 1} \right) \sqrt{1 + \left(\frac{x\left(3x^3 + 6x^2 + 4x + 2\right)}{\left(x^2 + 1\right)^2} \right)^2}\, dx$$

correct terminals and structure A1
correct solution A1

ii. total surface area $= 281.14\ldots + \pi f\left(\dfrac{3}{2}\right)^2 + \pi f(2)^2$

$= 808.4$ A1

Note: Consequential on answer to ***Question 1b.i.***

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c. **i.** $L = \int_{\frac{3}{2}}^{2} \sqrt{1 + \left(\frac{x\left(3x^3 + 6x^2 + 4x + 2\right)}{\left(x^2 + 1\right)^2} \right)^2}\, dx$

$= 5.08$ A1

ii. Using the non-rounded length from **part c.i.** gives:

$$\int_{2}^{q} \sqrt{1 + \left(\frac{x\left(3x^3 + 6x^2 + 4x + 2\right)}{\left(x^2 + 1\right)^2} \right)^2}\, dx = 5.08\ldots$$

$q = 2.34$ A1

Note: Consequential on working to ***Question 1c.i.***

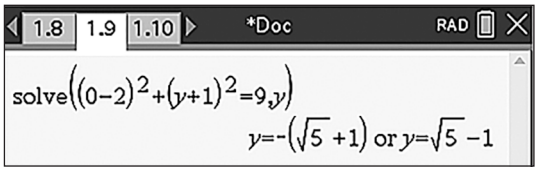
d. $V = \pi \int_{0}^{\frac{1}{2}} \left(\frac{x^5 + x^2 - 2}{x^2 - 1} \right)^2 dx$ M1

$= 6.89$ A1

Question 2 (11 marks)

a. The Cartesian equation of the relation is given by $(x - 2)^2 + (y + 1)^2 = 9$. M1

Using a CAS calculator to find the y-intercepts of the relation gives:



The y-intercepts are $\left(0, -1 + \sqrt{5}\right)$ and $\left(0, -1 - \sqrt{5}\right)$. M1

Therefore, finding the distance between the points of intersection gives:

$\text{distance} = -1 + \sqrt{5} - \left(-1 - \sqrt{5}\right)$

$= 2\sqrt{5}$ A1

b. The region bounded by the relation and the imaginary axis is a minor segment.

Using the cosine rule to find the angle subtended by the segment gives:

$\left(2\sqrt{5}\right)^2 = 3^2 + 3^2 - 2 \times 3 \times 3\cos(\theta)$

$\theta = \cos^{-1}\left(-\frac{1}{9}\right)$ M1

$\text{area} = \frac{r^2}{2}(\theta - \sin(\theta))$

$= \frac{3^2}{2}\left(\cos^{-1}\left(\frac{1}{9}\right) - \sin\left(\cos^{-1}\left(\frac{1}{9}\right)\right)\right)$

$= 3.10$ A1

Note: Consequential on answer to ***Question 2a.***

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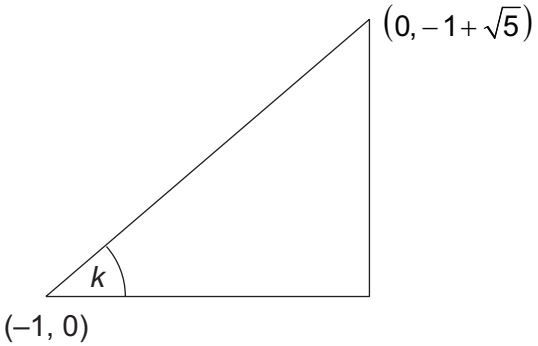
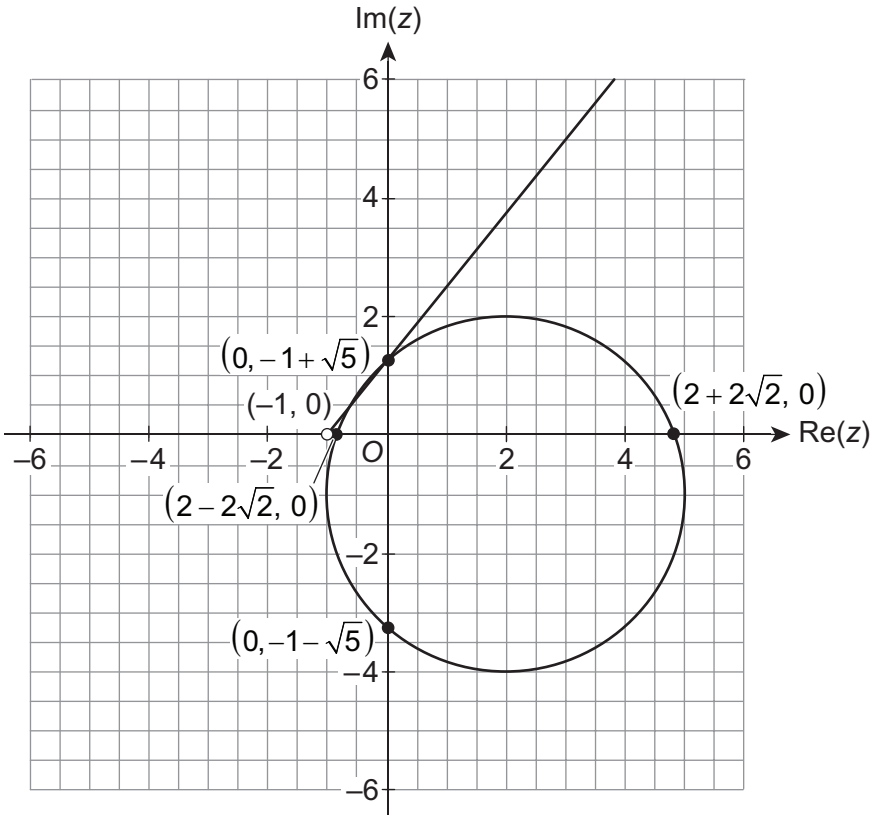
c. Using a diagram to find angle k gives:



$$\tan(k) = \frac{-1+\sqrt{5}}{1}$$ M1

$$k = 0.89^c$$ A1

d.



correct shape A1
correct intercepts A1

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e. The diameter of the circle is the line between points $-1 + 0i$ and $0 + \left(1 + \sqrt{5}\right)i$, and its centre is the midpoint of this line.

$$\text{diameter} = \sqrt{1^2 + \left(1 + \sqrt{5}\right)^2}$$
$$= \sqrt{2\sqrt{5} + 7}$$

$$\text{radius} = \frac{\sqrt{2\sqrt{5} + 7}}{2}$$

Therefore:

$$m = \left(\frac{\sqrt{2\sqrt{5} + 7}}{2}\right)^2$$
$$= \frac{2\sqrt{5} + 7}{4}$$ A1

$$\text{midpoint} = \left(\frac{-1 + 0}{2}, \frac{0 + \left(1 + \sqrt{5}\right)}{2}\right)$$
$$= \left(-\frac{1}{2}, \frac{1 + \sqrt{5}}{2}\right)$$

Therefore, $h = \frac{1}{2}$ and $k = -\frac{1 + \sqrt{5}}{2}$. A1

Thus, the cartesian equation for the circle is $\left(x + \frac{1}{2}\right)^2 + \left(y - \frac{1 + \sqrt{5}}{2}\right)^2 = \frac{2\sqrt{5} + 7}{4}$.

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Question 3 (8 marks)

a. $P = \int 2t^2 \cos(t)dt$

Let $u = 2t^2$ and $\frac{dv}{dt} = \cos(t)$.

Therefore, $\frac{du}{dt} = 4t$ and $v = \sin(t)$.

$P = 2t^2 \sin(t) - \int 4t \sin(t)dt$ M1

Let $u_2 = 4t$ and $\frac{dv_2}{dt} = \sin(t)$.

Therefore, $\frac{du_2}{dt} = 4$ and $v_2 = -\cos(t)$.

$P = 2t^2 \sin(t) - \left(-4t\cos(t) - \int -4\cos(t)dt\right)$

$= 2t^2 \sin(t) + 4t\cos(t) - \int 4\cos(t)dt$

$= 2t^2 \sin(t) + 4t\cos(t) - 4\sin(t) + c$ A1

b. **i.** $P(0) = 60$

$c = 60$

$P(5) = 21.6$ A1

ii. Substituting $t = 6$ into the rule for $\frac{dP}{dt}$ gives:

$\frac{dP}{dt} = 2 \times 6^2 \times \cos(6)$

$= 69.1$ cents per month A1

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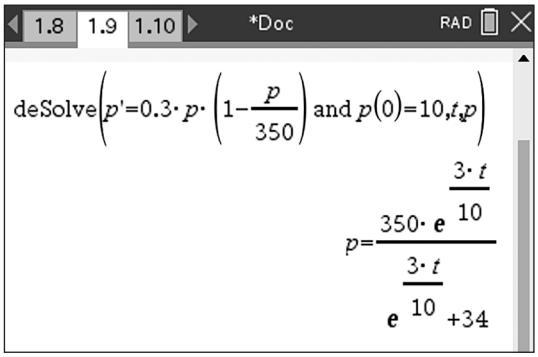
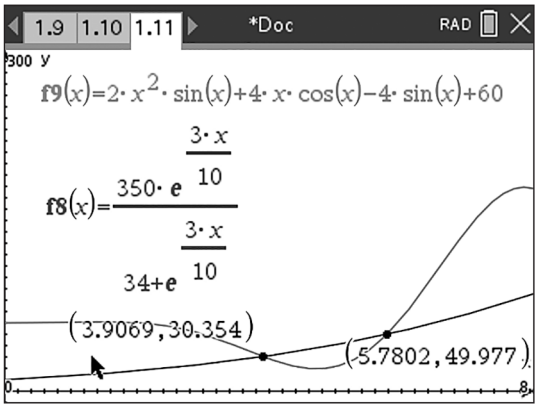
c. Using a CAS calculator to solve the differential equation gives:



$$P_B = \frac{350e^{\frac{3t}{10}}}{34 + e^{\frac{3t}{10}}}$$ A1

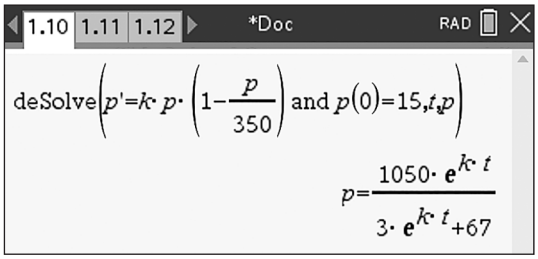
Using a CAS calculator to sketch the graphs of the functions on one set of axes gives:



The points of intersection show that the price of BBMT is greater than the price of AAMT between $t = 3.90\ldots$ and $t = 5.78\ldots$. M1

Therefore, BBMT is greater than AAMT for $5.78\ldots - 3.90\ldots = 1.87$ months. A1

d. Using a CAS calculator to solve the differential equation gives:



$$P_C = \frac{1050e^{kt}}{3e^{kt} + 67}$$

The graph of the price of CCMT must pass below the minimum value of the graph of the price of AAMT.

minimum values of AAMT $= 4.71 \ldots, 19.58 \ldots$

Substituting in $t = 4.71$ ... and $P_C = 19.58$ ... to find k gives:

$$19.58 \ldots = \frac{1050e^{k \times 4.71\ldots}}{3e^{k \times 4.71\ldots} + 67}$$

$$k = 0.05954 \ldots$$

$$= 0.059$$ A1

Note: Responses must be rounded down to 0.059 to ensure that the graph of CCMT passes below the minimum value of AAMT.

Question 4 (10 marks)

a. $x = 10\sin(t)$

$$t = \sin^{-1}\left(\frac{x}{10}\right)$$

$$y = 2\sin(2t)$$

$$= 2\sin\left(2\sin^{-1}\left(\frac{x}{10}\right)\right)$$ M1

$$= 4\sin\left(\sin^{-1}\left(\frac{x}{10}\right)\right)\cos\left(\sin^{-1}\left(\frac{x}{10}\right)\right)$$

$$= \frac{x\sqrt{100 - x^2}}{25}$$

As this represents only half of the relation, $y = -\dfrac{x\sqrt{100 - x^2}}{25}$.

Squaring both sides of the equation gives:

$$y^2 = \frac{x^2\left(100 - x^2\right)}{625}$$ A1

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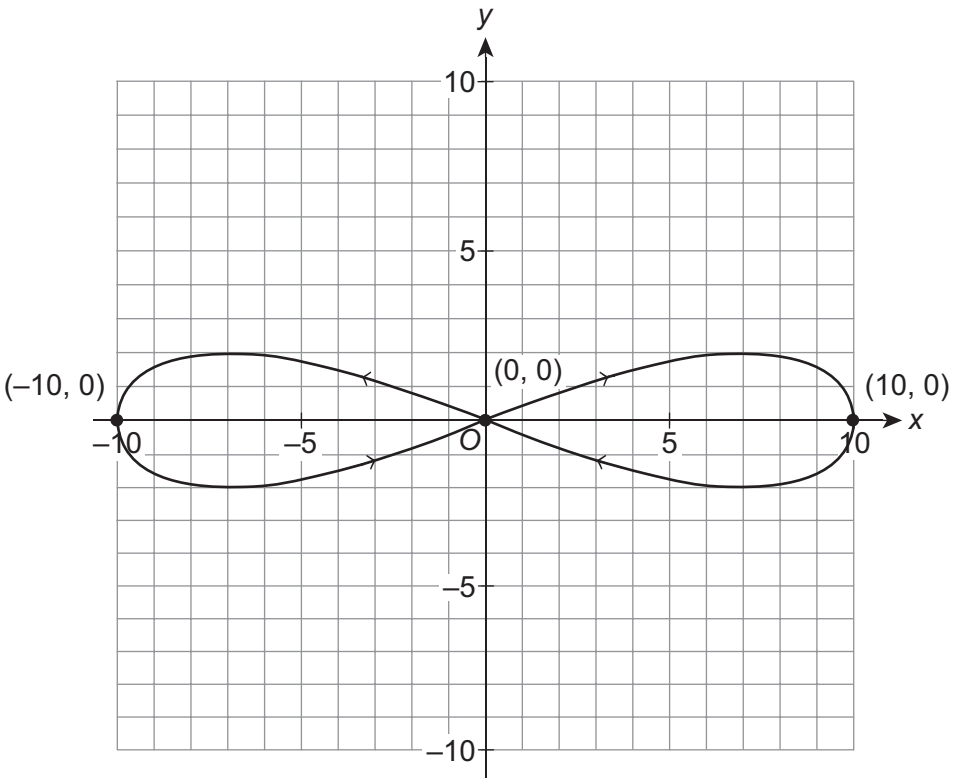
b.

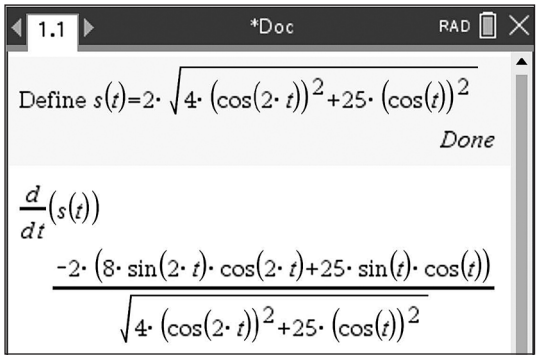


correct shape and direction of motion A1
correct intercepts A1

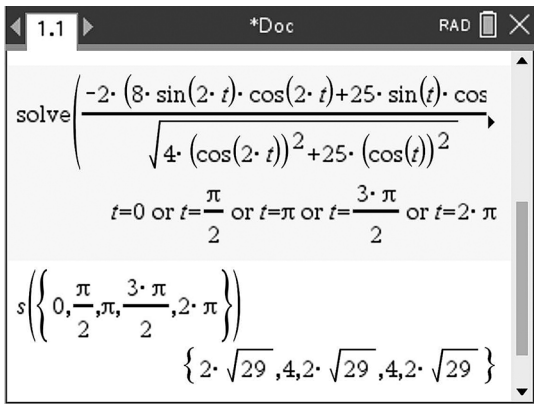
c. **i.** $|\dot{\underset{\sim}{r}}_c| = \sqrt{(10\cos(t))^2 + (4\cos(2t))^2}$

$= 2\sqrt{4\cos^2(2t) + 25\cos^2(t)}$ A1

ii. Using a CAS calculator to find the derivative of $\left|\dot{\underset{\sim}{r}}_c\right|$ gives:



The maximum and minimum speeds occur when the derivative $= 0$.
Therefore, using a CAS calculator to solve $\left|\dot{\underset{\sim}{r}}_c\right| = 0$ for t and hence the maximum speed gives:



maximum speed $= 2\sqrt{29}$ m s^{-1} A1

Note: Consequential on answer to ***Question 4c.i.***

iii. From working for **part c.ii.**:

minimum speed $= 4$ m s^{-1} A1

times at which minimum speed occurs $= \frac{\pi}{2}, \frac{3\pi}{2}$ s A1

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d. Finding the times at which the car passes through each component of point Q gives:

$$2\sin(2t) = -2$$
$$t = \frac{3\pi}{4}, \frac{7\pi}{4}$$
$$10\sin(t) = -5\sqrt{2}$$
$$t = \frac{5\pi}{4}, \frac{7\pi}{4}$$

Thus, the car is at point Q when $t = \frac{7\pi}{4}$.

Therefore, it takes the rock $\frac{7\pi}{4}$ s to fall s m. M1

Using the constant acceleration formula gives:

$$s = ut + \frac{1}{2}at^2$$
$$= 0 \times \frac{7\pi}{4} + \frac{1}{2} \times 9.8 \times \left(\frac{7\pi}{4}\right)^2$$
$$= 148.11 \text{ m}$$ A1

Question 5 (12 marks)

a. Let $\underset{\sim}{d} = -\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}$ and $\underset{\sim}{n} = \underset{\sim}{k}$, where $\underset{\sim}{n}$ is a vector normal to the plane defined by the x- and y-axes.

Finding the angle between $\underset{\sim}{d}$ and $\underset{\sim}{v}$ gives:

$$\theta = \cos^{-1}\left(\frac{\underset{\sim}{d} \cdot \underset{\sim}{n}}{|\underset{\sim}{d}||\underset{\sim}{n}|}\right)$$
$$= \cos^{-1}\left(\frac{2}{\sqrt{6}}\right)$$
$$= 35.26\ldots$$ M1

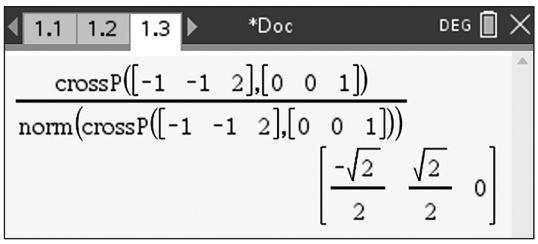
Therefore, the angle between L_1 and the plane is $90 - 35.26\ldots = 54.74°$. A1

b. Finding the shortest distance between L_1 and the z-axis gives:

$\left|\overrightarrow{PQ}\right| = \left|\left(\underset{\sim}{a}_2 - \underset{\sim}{a}_1\right) \cdot \hat{\underset{\sim}{n}}\right|$, where $\hat{\underset{\sim}{n}} = \dfrac{d_1 \times d_2}{\left|d_1 \times d_2\right|}$, d_1 and d_2 are vectors that are parallel to each line, and a_1 and a_2 are the position vectors of any two points, one on each line.

Hence, $\underset{\sim}{d}_1 = -\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}$, $\underset{\sim}{d}_2 = \underset{\sim}{k}$, $\underset{\sim}{a}_2 = \underset{\sim}{0}$ and $\underset{\sim}{a}_1 = 3\underset{\sim}{i} + 4\underset{\sim}{j}$.

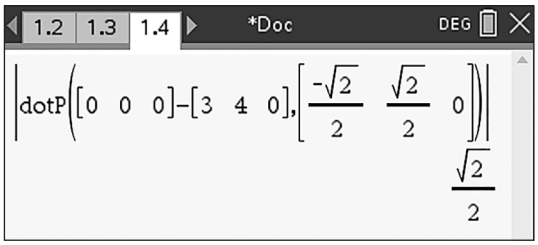
Using a CAS calculator gives:



$\hat{\underset{\sim}{n}} = -\dfrac{\sqrt{2}}{2}\underset{\sim}{i} + \dfrac{\sqrt{2}}{2}\underset{\sim}{j}$ M1

Therefore, $\left|\overrightarrow{PQ}\right| = \left|\left(\underset{\sim}{0} - \left(3\underset{\sim}{i} + 4\underset{\sim}{j}\right)\right) \cdot \left(-\dfrac{\sqrt{2}}{2}\underset{\sim}{i} + \dfrac{\sqrt{2}}{2}\underset{\sim}{j}\right)\right|$.

Using a CAS calculator gives:



distance $= \dfrac{\sqrt{2}}{2}$ A1

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c. Finding the equation of the plane containing A, B and C gives:

$\overrightarrow{AB} = 3\underset{\sim}{i} - 2\underset{\sim}{k}$, $\overrightarrow{AC} = 9\underset{\sim}{i} + 3\underset{\sim}{j}$

$\underset{\sim}{n} = \overrightarrow{AB} \times \overrightarrow{AC}$

$s = 6\underset{\sim}{i} - 18\underset{\sim}{j} + 9\underset{\sim}{k}$

Therefore, $6x - 18y + 9z = k$.

Substituting point A into the equation to find k gives:

$0 - 0 + 27 = k$

$k = 27$

Therefore, $6x - 18y + 9z = 27$. M1

Finding the angle between the normal vectors $6\underset{\sim}{i} - 18\underset{\sim}{j} + 9\underset{\sim}{k}$ and $\underset{\sim}{j}$ gives:

$$\theta = \cos^{-1}\left(\frac{\left(6\underset{\sim}{i} - 18\underset{\sim}{j} + 9\underset{\sim}{k}\right) \cdot \underset{\sim}{j}}{\sqrt{6^2 + 18^2 + 9^2}}\right)$$ M1

$$= \cos^{-1}\left(\frac{-18}{21}\right)$$

$$= 148.99\ldots$$

Therefore, the acute angle between the planes is $180 - 148.99\ldots = 31.0°$. A1

d. *For example:*

For the plane to be parallel to L_1, the vector normal to the plane must be perpendicular to a vector parallel to L_1.

Let the normal to the plane be $\underset{\sim}{n} = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k}$. Therefore, $-x - y + 2z = 0$. M1

There are infinitely many values that satisfy this equation; one solution set is $x = 1$, $y = 1$, $z = 1$.

Therefore, the cartesian equation of a plane that is parallel to L_1 that goes through the origin is $x + y + z = 0$. A1

Note: Accept any correct cartesian equation.

e. Finding the shortest distance between a point on the plane, such as the origin, and L_1 gives:

$-(3 - t) - (4 - t) + 2(2t) = 0$

$t = \frac{7}{6}$ M1

Substituting $t = \frac{7}{6}$ into the original equation for L_1 gives the point $\left(\frac{11}{6}, \frac{17}{6}, \frac{7}{3}\right)$.

The distance between $\left(\frac{11}{6}, \frac{17}{6}, \frac{7}{3}\right)$ and the origin is $\frac{\sqrt{606}}{6}$. A1

f. *For example:*

The plane must be parallel to the plane $x + y + z = 0$; therefore, $x + y + z = k$.

Using the point (3, 4, 0), $k = 7$.

Therefore, the cartesian equation of a plane that contains L_1 is $x + y + z = 7$. A1

Note: Accept any correct cartesian equation.

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Question 6 (9 marks)

a. H_0: $\mu = 700$

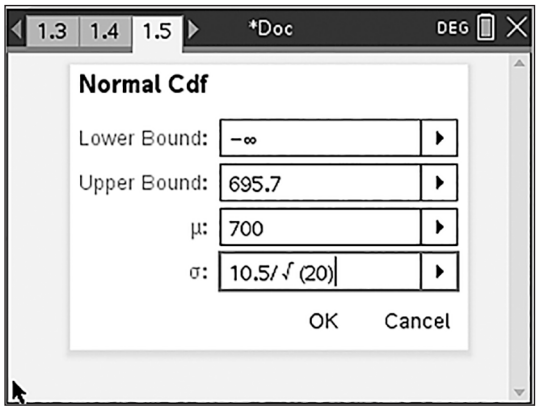
H_1: $\mu < 700$

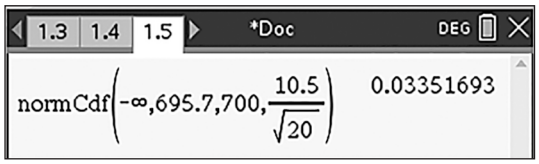
A1

b. **i.** $p \text{ value} = \Pr\left(\bar{X} < 695.7 \mid \mu = 700\right)$

$s = \dfrac{10.5}{\sqrt{20}}$

Using a CAS calculator gives:





p value $= 0.034$ A1

ii. Yes, as the p value is less than 0.05. A1

c. $C = A + B$

$\text{E}(C) = \text{E}(A) + \text{E}(B)$

$= 300 + 400$

$= 700$

$\text{Var}(C) = \text{Var}(A) + \text{Var}(B)$

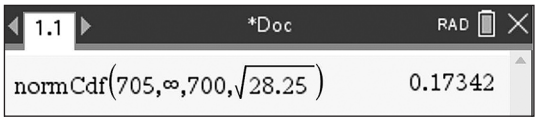
$= 3.5^2 + 4^2$

$= 28.25$ M1

$\sigma(C) = \sqrt{28.25}$

$= 5.31\ldots$

Using a CAS calculator gives:



Therefore, $\Pr(C > 705) = 0.173$. A1

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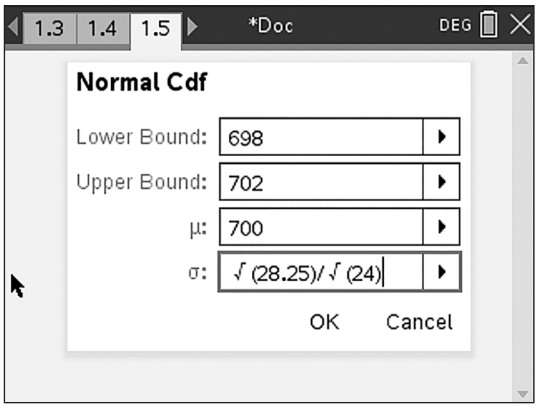
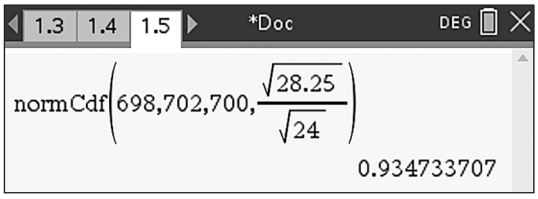
d. $E(\bar{X}) = 700$

$\sigma(\bar{X}) = \dfrac{\sqrt{28.25}}{\sqrt{24}}$ M1

Using a CAS calculator gives:





Therefore, $\Pr(698 < \bar{X} < 702) = 0.935$. A1

e. $M \sim N\left(700, 10.5^2\right)$ and $C \sim N\left(700, \sqrt{28.25}^2\right)$

$Q = M - C$

$\text{E}(Q) = \text{E}(M) - \text{E}(C)$

$= 700 - 700$

$= 0$

$\text{Var}(Q) = \text{Var}(M) + \text{Var}(C)$

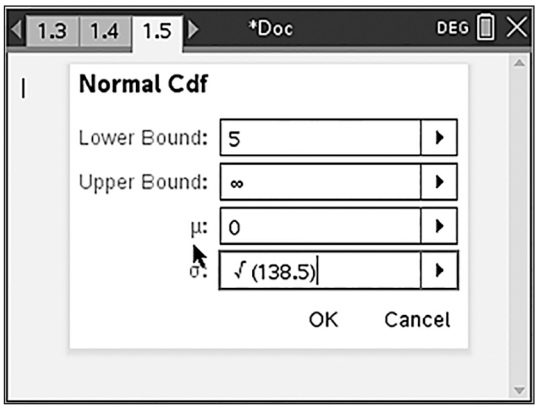
$= 10.5^2 + 28.25$

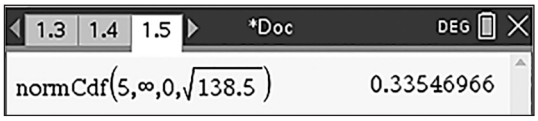
$= 138.5$

$\sigma(Q) = \sqrt{138.5}$

$= 11.76\ldots$ A1

Using a CAS calculator gives:





$\Pr(Q > 5) = 0.34$ A1

Note: Consequential on working to ***Question 6c.***

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